

Colin Murphy

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PROFESSIONAL SUMMARY

Founding engineer and full-stack builder with 8+ years building end-to-end AI products for sensitive, data-intensive industries. Repeatedly the sole engineer behind LLM-powered healthcare platforms — owning the full stack from React/Next.js frontends through Python LLM services and EHR/FHIR pipelines to cloud infrastructure (AWS & GCP), with deep strengths in LLM evaluation and grounding for domain-specific accuracy.

TECHNICAL SKILLS

- **AI & Machine Learning:** LLMs (Fine-tuning, RAG, Evaluation, LangChain, Langfuse), Computer Vision, Deep Learning (PyTorch, TensorFlow), Model Explainability.
- **Full Stack & Prototyping:** TypeScript, React, Next.js (App Router), Python, FastAPI, gRPC, Streamlit, Docker, CI/CD.
- **Data Engineering:** Unstructured & Clinical Data Pipelines (PDF Parsing/OCR, FHIR/EHR), SQL, NoSQL, Big Data Management, ETL.
- **Cloud & Infrastructure:** AWS & GCP (Cloud Run, Firestore, Pub/Sub), MLOps, Model Serving, Model Lifecycle Management.

WORK EXPERIENCE

- **PACT** New York, NY
Full-Stack AI Engineer (Contract) 03/2026 – 06/2026
 - Sole engineer maintaining and extending a production LLM platform that matches patients to clinical trials, spanning a Next.js frontend and two Python backend services.
 - **Full-Stack Delivery:** Shipped patient-facing features across a Next.js 14 (App Router) / React / TypeScript frontend and gRPC + FastAPI Python services, owning the full path from UI to LLM matching engine to EHR data layer.
 - **LLM Matching & Evaluation:** Improved the clinical-trial eligibility pipeline (litellm / OpenAI), building criteria-level evaluation harnesses and resolving structured-output failures to raise matching reliability.
 - **EHR Integration (FHIR):** Integrated FHIR R4 patient records via the Fasten Health API, normalizing clinical data into PostgreSQL for downstream matching.
 - **Cloud & Reliability:** Operated the platform on GCP (Cloud Run, Firestore, Pub/Sub); diagnosed production incidents and hardened async matching against redelivery and race-condition failures, all under PHI-safe logging.
- **HealthyMe AI** New York, NY
Founding Engineer / Head of Engineering 10/2023 – 02/2026
 - Served as the sole engineer scaling technology from zero-to-MVP; currently leading a team of 4 to build next-gen decision support tools.
 - **LLM & Reasoning Pipelines:** Developed LLM pipelines using RAG and LangChain. Grounded model outputs in domain literature to provide accurate, well-supported answers for high-stakes clinical decision-making.
 - **Unstructured Data Engineering:** Built ingestion pipelines to parse, OCR, and extract insights from over 100,000+ unstructured PDF patient reports and 9 million images, transforming raw data into a structured knowledge base.
 - **Data Annotation Infrastructure:** Designed and built the core web platform for data acquisition, establishing the database structures required to coordinate inputs from a distributed team of global experts.
 - **SOTA Model Development:** Trained a custom classification model capable of predicting over 60 distinct dermatological diseases. Achieved performance metrics surpassing current best-in-class numbers in peer-reviewed literature.
- **Pall Corporation** New York, NY
Machine Learning Engineer 01/2022 – 08/2023
 - Established model lifecycle and deployment scheme for machine learning applications, moving research prototypes into live workflows.
 - Developed computer vision tools for automated characterization, moving R&D concepts into manufacturing reality.
- **Johnson & Johnson** West Chester, PA
Scientist / Data Scientist 08/2016 – 12/2021
 - Led an interdisciplinary team applying data science to complex manufacturing systems.
 - Built time-series and forecasting models to predict process failures and optimize high-value manufacturing decisions from semi-structured data.

EDUCATION

- **Drexel University** Philadelphia, PA
M.Sc. Data Science 06/2022
- **Temple University** Philadelphia, PA
B.Sc. Chemistry | Phi Beta Kappa 05/2016